

✓ Pricing commodities – coffee

Using the provided market data, apply the cost of carry model, the Black-Scholes model, and Monte Carlo simulations to determine the fair price of a coffee option contract. Analyze how changes in supply, demand, weather, and geopolitical factors might impact your pricing.

Given:

- Current spot price - S_t : \$1.20 per pound
- Risk-free rate - r : 2% per annum
- Storage cost - d : 1% per annum
- Time to maturity - T : 6 months (0.5 years)
- Strike price - X : \$1.25
- Volatility - σ : 25%

```
import numpy as np
from scipy.stats import norm
```

```
#input given data
St = 1.20
r = 0.02
d = 0.01
T = 0.5
X = 1.25
sigma = 0.25
num_simulations = 10000
num_steps = 252
dt = T / num_steps
```

✓ Cost of Carry Model

```
Ft = St * np.exp((r+d)*T)
print(f"The fair price of the coffee futures contract is ${Ft:.3f} per pound.")
```

The fair price of the coffee futures contract is \$1.218 per pound.

✓ The black Scholes Model

```
# Calculating d1 and d2
d1 = (np.log(St / X) + (r + 0.5 * sigma ** 2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)

# Calculating call option price using Black-Scholes formula
C = St * norm.cdf(d1) - X * np.exp(-r * T) * norm.cdf(d2)
```

```
print(f"The price of the call option is ${C:.3f}.")
```

The price of the call option is \$0.068.

✓ Monte Carlo simulation

```
# Simulating price paths
np.random.seed(42) # For reproducibility
price_paths = np.zeros((num_steps, num_simulations))
price_paths[0] = S_t

for t in range(1, num_steps):
    z = np.random.standard_normal(num_simulations)
    price_paths[t] = price_paths[t-1] * np.exp((r - 0.5 * sigma ** 2) * dt + sigma * z)

# Calculating the average simulated price at maturity
average_simulated_price = np.mean(price_paths[-1])
print(f"The average simulated price of the coffee futures contract at maturity is ${average_simulated_price:.3f}.
```

The average simulated price of the coffee futures contract at maturity is \$1.210.